

Sensors and Automated Filtration Systems for Betta Fish

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Abstract— Betta fish are well known for their unique yet elegant vibrant colors, particularly their fin types. But eccentric features come with a toll that is extra care for their living space. Requirements for a healthy and clean water environment to survive is essential because exposure to poor water quality are prone to stress, diseases and death. Maintaining betta fish could be exhausting. Pet enthusiasts with busy work hours often neglect basic procedures such as manual tank cleaning. This paper provides a prototype development of an automated and smart filtration system with a turbidity sensor to detect water quality dirtiness level in real time. The filtration system uses reverse osmosis (RO) to filter out the water and is capable of cleaning the fish tank with ease without manual intervention, ensuring the fish tank remains in a clean state automatically. Test results show that the device is working properly as an automated filtration device. The proposed design provides a step to reach the future of IoT implementation by owning pets. Further research is needed to improve the proposed prototype design towards a smarter system.

Keywords—Betta fish, filtration system, IoT, reverse osmosis, water environment

I. INTRODUCTION

In Indonesia, it is common for people to live alongside their pets. According to a survey conducted by Litbang Kompas in January 2020, the five most kept animals in the Greater Jakarta area (*Jabodetabek*) include fish, cats, dogs, birds, and rabbits.

Among those pets, Betta fish, also known as Siamese fighting fish—are particularly popular due to their vibrant colors, elegant fin shapes, and high market value and requests [1], [2]. However, maintaining a Betta fish requires special care, especially in terms of water quality. Many Betta fish owners are still struggling to provide proper water conditions, specifically amateurs or even new owners, which are crucial for the fish's health and longevity [2].

As technology becomes more advanced and standardized every day, the usage of technologies like a simple automation device can be implemented to help pet owners, and in this case, Betta fish. Implementation of microcontrollers like Arduino could help the process of taking care of the Betta fish.

Maintaining extra care for Betta fish isn't as easy as it seems; personal matters could affect how Betta fish owners maintain their own pet. In other words, to maintain such care towards Betta fishes, there must be automated devices so that it could help owners take care of their Betta fishes and clean the aquarium [1]

II. THEORITICAL BASIS

A. Fishtank Clean Water

Clean water means it is clean from bacteria, viruses, and sediments[3]. If the water is proven to contain toxic matter, it is indicated that it is spoiled [4]. To achieve the term clean water, there must be visual observation along with the test of water content such as taste, color, smell, and the water clarity [5]. Another standard or parameter that is used is microorganisms and chemicals like lead or copper.

The usage of water inside an aquarium is the same as using water for daily human consumption, where small mistakes can have such a fatal impact. Small changes in the aquarium water like chemicals could change the whole water. Just like humans are sensitive to pollution, fish like Betta fish are also sensitive to unwanted pollution inside the aquarium [1]. Because it will impact the health of the Betta fish and other living organisms inside the aquarium. Not only will the fish get clean water, but the owners will also be able to enjoy the clean scenery of the aquarium.

B. Betta Fish

Betta fish (*Betta splendens*) or as Indonesians say it, *Cupang* is known as the Siamese fighting fish. It is also one of the most popular tropical freshwater fish in the world due to its color along with its fins and body shape and relatively easy to take care of. Betta fish have over 70 types of species, each with distinct colors, patterns, and fin shapes.

Betta fish need air from inside and on the surface of the water. Water fillings should not be careless because it is to prevent the fish from jumping out of the aquarium. However, betta fish can survive in water with a small volume and low oxygen, even with stagnant water. A larger aquarium is the better standard for betta fish maintenance and health. Betta fish aquariums should be at least 2.5 gallons in size, with the recommended size being 5 gallons[2] for the long term.

Betta fish are carnivorous fish, so they need more high-protein foods. The recommendation for betta fish is high-protein fish pellets that usually contain krill, shrimps, and worms[2]. However, it should be noted that overfeeding Betta fish can have a negative impact on the water quality and the surrounding environment because leftover food will accumulate at the bottom of the aquarium.

C. Filtration System

Water filtration means separating contaminants like chemicals and sediments from the water. Filtration consists of four stages in which each stage uses different types of filters to filter out sediments, chemicals, impurities [6], and

lastly to remove the remaining bacteria or microorganism that might live in the water using ultraviolet light or ozone treatment.

D. Water Sensors

The water sensor is used to measure turbidity in the water. It works by taking measurements of the water clarity by analyzing lights. If the lights are scattered in the water, it means there is something that obstructs the light, either sediments, dirt, or other things that may block the light into the sensor [7]. This also includes organic matters too like plants or algae. High turbidity readings mean it's not clear and potentially harmful. Especially for the Betta fish to live.

E. Reverse Osmosis

A process that occurs when water is forced through a semi-permeable membrane and directly separates water from unwanted substances [8] such as contaminants, sediments, and impurities. The rejected water will be effectively removed and purified [9] through a staged filtration to later be solvent into clean water.

III. METHODOLOGY

A. System Overview

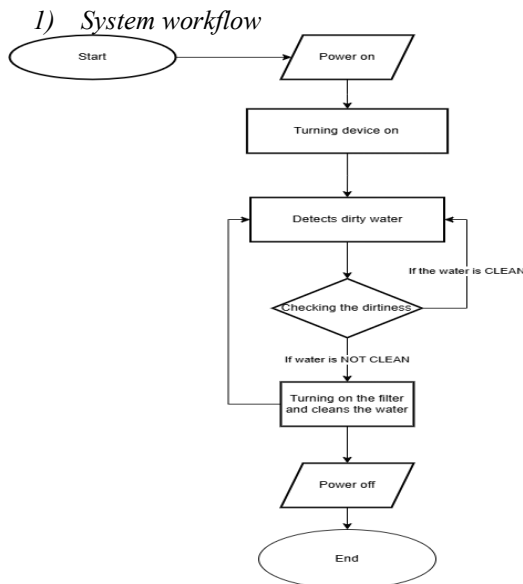


Fig. 1. System Flowchart

Fig 1. shows the system overall flow from the moment it is on and off. The device will first receive power from the battery or any kind of electric power. And then after the device is on, the turbidity sensor will detect water dirtiness for an interval of time. And it will check the water dirtiness, if it is dirty, the device will run the filter on and clean it. If the turbidity sensor does not detect any scattered lights [4], [7], it means it's not dirty and the device will re-check the water dirtiness.

2) Turbidity Process

Turbidity sensors distinctly measure the intensity of a liquid, based on their clarity, cloudiness, and haziness. Data that is received by the sensor must be scattered or random particles and then absorbed. Detecting the light that is transmitted will indicate the water readings. For instance, if the light is spread evenly throughout the displacement, this will indicate the water content is high turbidity scale. High turbidity readings mean the water is contaminated and

necessary to be treated [7] for further notice. Materials that can trigger these readings are substances that are opaque or impenetrable such as clay, dirt and other compounds.

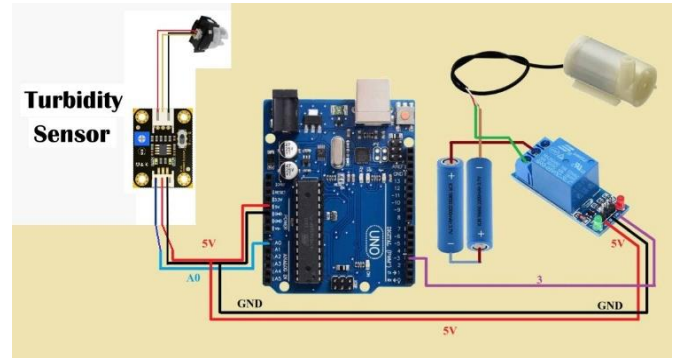


Fig. 2. Turbidity sensor circuit

Fig 2. illustrates how the component is built. All data that is received by the sensor must be displayed. Conducted experiments show the current readings of the water turbidity and expressed in Nephelometric Turbidity Units (NTU) [10]. These turbidity sensors aren't built up automatically. They need to contain a library for Arduino that allows control of the LCD (Liquid Crystal Display) modules via the I2C (Inter-Integrated Circuit) protocol by inserting "NewLiquidCrystal" library. This library improves the I2C, SPI, and parallel interfaces support.

Turbidity sensor that is measured that needs relative light transmission can be defined by the absorbance logarithmic value

$$A = \log(\theta_0 / \theta_i) [11]$$

θ_0 is the incident luminous intensity and θ_i is the transmitted luminous intensity. As a consideration that:

- (1) The output voltage (V_{out}) of the sensor is inversely proportional to turbidity, so,
- (2) The clear water has a close value to 0% turbidity, and
- (3) Turbidity of 100% means a fully opaque liquid.

The equation for turbidity in percentage units:

$$\text{Turbidity} = 100 - (V_{out}/V_{clear}) * 100$$

3) Relay Implementation

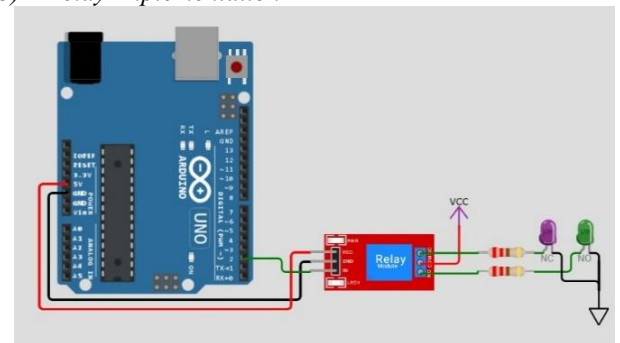


Fig. 3. Relay circuit

This turbidity sensor also will trigger the relay layout to operate the water pump actuators as it is shown in Fig 3. The connections are between the aquarium and the reverse osmosis filter. The threshold logic in this experiment is based on turbidity levels measured by the sensor. When the turbidity value exceeds 10, the water is considered dirty. In response, the microcontroller activates the relay, which turns on the pump to either filter or drain contaminated water. During this state of mind, the LCD display shows the

message “PUMP ON” to indicate that the system is actively managing water quality. Conversely, when the turbidity value is 10 or below, the water is considered clean. The relay is then deactivated, turning the pump off, and the LCD updates to show “PUMP OFF.” Throughout the process, real-time turbidity readings and pump status are continuously displayed on both the LCD screen and the Serial Monitor, providing clear and immediate feedback to the user. It indicates that the water is safe and clear.

4) ESP32 Implementation

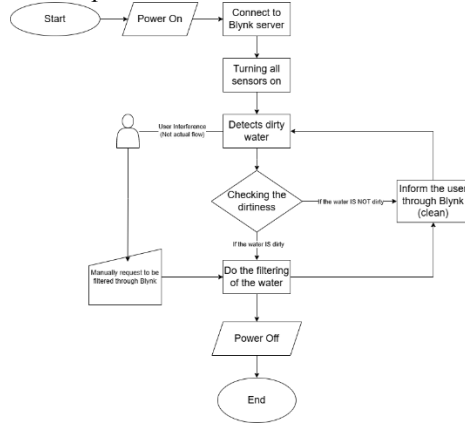


Fig. 4. Proposed ESP32 Implementation

Future implementation in IoT that can be implemented is using ESP32 instead of Arduino R3. ESP32 can be figured out and programmed using the same software as Arduino R3, which is Arduino IDE. Similar with Arduino R3, ESP32 also has many GPIO pins [12]. Making it compatible with the previous use of Arduino R3 as the microcontroller of the whole system with additional Wi-Fi module in it [13]. The analog signal received from the turbidity sensor will be forwarded over the internet, unlike previous prototype with Arduino R3.

Fig 4. shows that by using ESP32 to send information over the internet through the Blynk app, users can control and interfere the filtration timing despite the automation of it. Since the Blynk app supports IoT and ESP32 [13], displaying information would be a lot more beneficial if the data was interpreted into real-time information for the user.

5) Scikit-Learn's 'StandardScaler'

Standardizing a dataset is often necessary in machine learning because many models perform poorly when the features differ greatly in scale or distribution, rather than being roughly like a standard normal distribution. 'StandardScaler' in Scikit-Learn helps to ensure that the model appropriately weighs each feature [14]. It is common to have different features to be significantly different scale, it fits data and transforms to standard [14] distribution where each feature's mean = 0. On handling this data, training and testing should remain separated which means never scale the whole dataset but transform some testing data. Dataset that are featured should allow interaction and predictions [14].

6) Reverse Osmosis

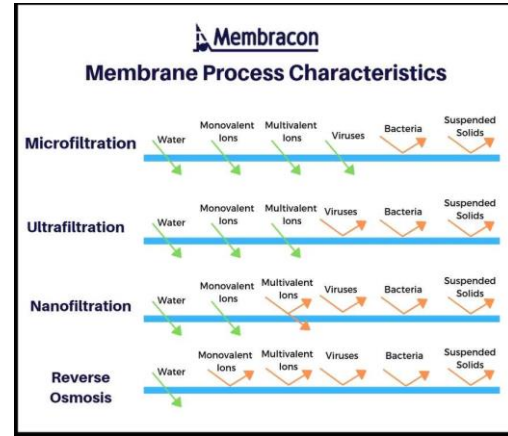


Fig. 5. Membrane Process Characteristics

The turbidity cycle relay technology integrated directly with the reverse osmosis process method. Utilizing semi permeable membrane to separate water substances and contaminants [15]. Note that the cycle will be operative when the relay is activated by the turbidity sensor. The process applied to treat wastewater since normal treatment programs as shown in the table below do not remove dissolved solids, the advantage of RO process characteristics is to remove and separate solid material also chemical pollutants [16] present in wastewater. Fig 5. shows how reverse osmosis filtrates most of the components in the water. The RO process consists of four major systems [8]:

- Pretreatment system
- High pressure pumps
- Membrane systems
- Post-treatment

7) Difference Between Forward and Reverse Osmosis

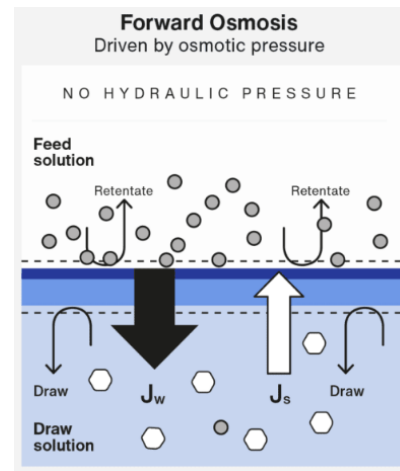


Fig. 6. Forward Osmosis

Fig 6. illustrates how forward osmosis works, osmotic pressure flow of water through the membrane at a relatively low pressure which eliminates the need for high-pressure hydraulics [17].

Systematic of the FO process:

- Optimize areas used with simpler and efficient treatment and require less placements.
- Reuse more water with better quality by treating streams that usually are difficult to contaminate.
- Low pressure [18] application makes the FO membrane less sensitive to scaling and fouling

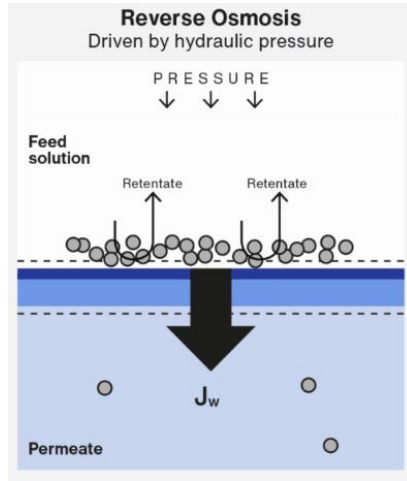


Fig. 7. Reverse Osmosis

Fig 7. illustrates how reverse osmosis is done, water that is forced through the membrane that utilizes pressure to overcome natural pressure.

Systematic of the RO process:

- Semi-permeable membrane [19] to separate water molecules from other substances.
- The RO membrane is identified as filter, flowing water molecules to reject contaminants [20].
- High pressure [21] application makes the RO membrane force water across the system.

8) Filtration Contents

The reverse osmosis process will not be clean unless the contents fulfill the filtration cycle consisting of mechanical filtration canister. An effective canister chamber is necessary to sustain a habitat which is used to provide a place to contain good bacteria. This filtration chamber and function consists of:

- Cotton: acts as a barrier [22], primarily removing larger particles and sediment from water.
- White coral: water filtration by releasing microbes [23] into the water column.
- Seashells: remove heavy metals and other pollutants, for controlling high calcium carbonate [24] content.
- Lava rock: provides a large area for beneficial bacteria to colonize and grow [25].

IV. RESULT AND ANALYSIS

TABLE I. TABLE DATA

Dirt Amount (Tablespoons)	Water Status		
	Water Cloudiness Level	Time to Clean Water (Minutes)	Clarity After RO Filtration
1 tbsp	Cloudy	11:57.51	Clear
2 tbsp	Cloudy	18:36.66	Clear
3 tbsp	Dirty	29:05.87	Clear
4 tbsp	Dirty	37:49:73	Clear
5 tbsp	Dirty	48:13:40	Clear

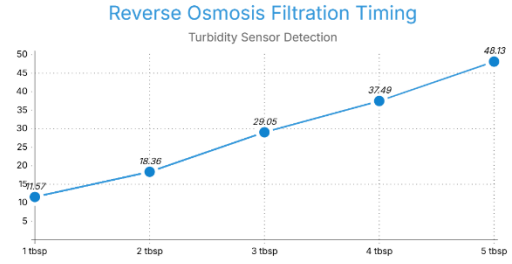


Fig. 8. Filtration time required by the RO system to reach acceptable clarity based on turbidity sensor data.

A. Experiment Observations

After conducting the experiment by inserting a certain amount of dirt into the turbidity cycle system measured in tablespoons, also measured by time recording taken required by the reverse osmosis system to restore the water clarity from debris, dirt and color evaluated completely. The data in Table 1 concludes the estimated filtration process time as the water being treated into clean water. It reveals the number of tablespoons of dirt increases so the time needed for treating dirty water also increases. Fig 8. illustrates the extent of the graph. The water was initially observed to be either *cloudy* or *dirty* based on the dirt load. The RO system was able to restore clarity in all cases, with the time to achieve turbidity threshold levels increasing with each additional tablespoon of dirt.

B. Filtration time and Dirt Quantity Analysis

Based on Table 1 and figure 7, a near linear increase in filtration time is observed as dirt content increases in volume. For instance, 1 tablespoon full of dirt results in a filtration time of 11:57.51 to 5 tbsp results in filtration time rising from approximately 12 minutes to over 48 minutes. This behavior suggests that although RO systems consistently produce clear water, their efficiency decreases significantly as the contamination load increases. This linearity also implies that turbidity sensors can serve as reliable indicators of required filtration duration in RO systems under particle-laden conditions.

C. System Efficiency and Implications

The turbidity sensor paired with reversed osmosis system can effectively remove visible contaminants across all testing conditions, confirming the reliability of the RO filtration is capable of handling low to moderate matter. Regardless of how the progressivity of cleaning time is observed, especially nodes of dirt with more than 3 tbsp of dirt, indicates that filtration system is influential for further treatment. Hence, the consistently clear water after filtration shows that the RO system integrity worked well and all membrane configured stayed intact, even with higher amounts of dirt. While longer filtration times can lead to some clogging or wear over time, maintenance is still important. However, it's worth noting that maintenance is not required as frequently as in conventional filtration systems, since RO systems are more efficient at handling fine particles and maintaining long-term performance.

D. Analysis and Future Implementations

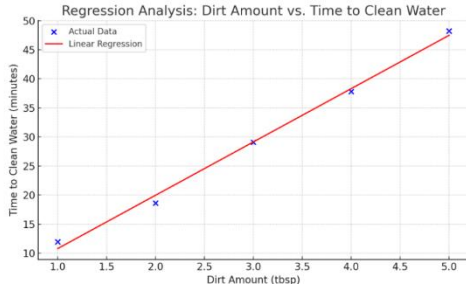


Fig. 8. Regression analysis for dirt amount vs time to clean water

Fig 8. shows the regression of dirt with respect to time, indicating that the dirtier the water gets, the longer it takes to filter out the water. Linear regression with the equation:

$$\text{Time} = 9.17 \times \text{Dirt Amount} + 1.62$$

Indicating from the equation above that positive slope 9.17 for each tablespoon added increases the cleaning time iteration by 9.17 minutes and with the intercept 1.62 as for no debris cleaning, as shown in Fig. 8, shows that the sensor accurately detects changes and water dirtiness.

The device works well as automated filtration. A possible future implementation may include IoT implementation and adding other sensors other than turbidity sensors to measure many more parameters. IoT implementation could be connecting the device with a wi-fi module or using ESP32 to send the data through the internet and to have it send a notification from the user's phone. This way, the device will send a notification when the water gets dirty. Making it easier for the user to take care of their pet, Betta fish.

Adding other sensors might help the user or pet's owner build a certain parameter for their pet. Some other sensors might be a pH sensor, temperature sensor, etc. Other than that, the goal is to improve the quality of the water tank for the Betta fish. Some ideas like adding other devices or even a smart AI tracking for fish movements, water quality, fish behavior, or perhaps AI use of predicting water results can be implemented to further improve the experience of taking care of the Betta fish, especially for new owners or amateurs.

V. PROPOSED AI IMPLEMENTATION

In this proposed AI implementation, using previous linear regression from Fig 8. means that it is a supervised learning environment [26]. The simple linear regression produces an equation as above:

$$\text{Time} = 9.17 \times \text{Dirt Amount} + 1.62$$

Dirt amount can be considered as the independent variable in the linear regression for supervised learning. Using regression analysis, this linear regression serves as datasets to train the model to predict the time it takes to clean or the dependent variable [26], [27]. The trained model will serve its prediction based on this supervised training, which is to predict the time it takes to clean.

AI for prediction purposes regression trained model will always provide good predictive [28] target filtration time sequence. Variables included are turbidity, temperature, pH, tank volume, filter age and initial TDS.

Every time filtration is conducted, system always records and track data such as timestamp, tds_start, turbidity_start, ph, tank_volume_l, flow_rate_lpm, filter_age_days,

target_tds, target_turbidity, and the measured time_to. This dataset is periodically exported for model training.

Linear Regression is accurate and faster for testing on microcontrollers and computational complexity. For larger datasets, lightweight tree-based models such as Gradient Boosting run on a smaller gateway. Recommended trained weights, standard deviations, bias, and scaler parameters [29] from the Ridge pipeline and hard-code them on the ESP32 utilize Python and scikit-learn [14].

Example training:

```
from sklearn.linear_model import Ridge
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline import pandas as pd
df = pd.read_csv("filtration_sessions.csv")
FEATURES =
["tds_start","turbidity_start","temperature","ph","tank_v
olume_l",
"flow_rate_lpm","filter_age_days","target_tds","target_t
urbidity"]
X = df[FEATURES]
y = df["time_to_target_min"]
model = Pipeline([("scaler", StandardScaler()), ("ridge",
Ridge(alpha=1.0))]) model.fit(X, y)
```

Doing this supervised learning system enables predictive control, reducing unnecessary pump operation, improving water quality consistency. It also allows the device to learn and adapt to changing condition or temperature changes.

VI. CONCLUSION

Both this paper and the device are an early step for the development of a fully detailed IoT device to be used for taking care of a Betta fish automatically. The tests showed that the device worked well. However, the current device is limited to only automation of filtration enough to help clean the water by using reverse osmosis on the filtration system. This solution provides simple help for the pet's owner and a significant step towards a smart and automated Betta fish care. Furthermore, proposed implementation such as predicting the time it takes to filter using machine learning prediction and simple linear regression from the data and using ESP32 for more IoT implementation could be added in future research for better application of the prototype and product.

AUTHOR CONTRIBUTION

S.Y. Purnomo, C.D.J. Kurnia, along with D.R. Hermanus contributed equally to this paper. S.Y. Purnomo handled the introduction, literature review, some part of the theoretical basis, and evaluation. C.D.J. Kurnia handled the theoretical basis and methodology along with testing the prototype. D.R. Hermanus supervised this research and provided critical revisions and suggestions. All authors contributed to the final discussion and result analysis

DATA AVAILABILITY

The results and the figures used to explain the process and analysis are available at this link:

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